21CTSTD2P4

SomeonecoolLovesMaths

May 31, 2026

Problem

Let $f(x), g(x)$ be two polynomials with integer coefficients. It is known that for infinitely many prime p, there exist integer m_p such that

$$f(a) \equiv g(a + m_p) \pmod{p}$$

holds for all $a \in \mathbb{Z}$. Prove that there exists a rational number r such that

$$f(x) = g(x + r).$$

Claim 1 — $\deg f = \deg g$.

Let $p > \max(\deg f, \deg g)$. Then, due to Lagrange's Theorem, $g(x + m_p) - f(x)$ is the zero polynomial in $\mathbb{F}_p[X]$. So the degree has to be equal along with the leading coefficients. $\square$

Now let the coefficient of x^{n-1} in f, g be a_{n-1} and b_{n-1} respectively. So we have that $a_{n-1} \equiv nm_pb_n + b_{n-1} \pmod{p}$. So, $m_p \equiv \dfrac{a_{n-1} - b_{n-1}}{nb_n} = r \pmod{p}$ for some rational number r.

Let the denominator of r be q, so, let $h(x) = q^n(g(x + r) - f(x))$. So, $p \mid h(x)$ for any large p, which implies that h is the zero polynomial.

Since $q \neq 0$, so, $f(x) = g(x + r)$ for some $r \in \mathbb{Q}$. $\square$